

AI-Assisted Triage: Data Exploration

Link to Github Notebooks: [Initial exploration and visualization \(notebooks\)](#)

One-sentence verdict

The ED triage dataset appears good enough to build a triage machine learning model, but important limits must be handled carefully and the model must be validated locally before it is used to support real patient-care decisions.

Dataset summary

- Dataset: `yaleemmlc\_admissionprediction\_triage.csv`
- The dataset contains 55,121 emergency department arrival records and 225 columns.
- The main target variable is ESI (Emergency Severity Index), the triage level assigned to each patient. ESI runs from 1 (most urgent) to 5 (least urgent).
- The data includes demographics (age, gender, race, ethnicity), arrival and triage details, vital signs (heart rate, blood pressure, respiratory rate, oxygen saturation, temperature, glucose) and 200 binary chief complaint columns (each starting with cc_).
- Most patients fall in the middle acuity levels: ESI 3 is the largest group at 27,010 patients (49.0%), followed by ESI 2 at 17,924 patients (32.5%). ESI 1 is very rare with just 77 patients (0.1%), meaning there are few examples for a model to learn from when it comes to the sickest patients.
- There are 0 missing ESI labels, so every patient has a recorded triage outcome to train against.

Top 3 data-quality concerns

1. Very few of the most urgent (ESI 1) patients

Only 77 patients (0.1%) are ESI 1. A model could look accurate overall while still doing a poor job identifying the sickest patients since it has so few examples to learn from. **Planned response:** In Week 6, check model performance separately for each ESI group, especially ESI 1 and ESI 2, so weak performance on the most urgent cases isn't hidden by strong overall accuracy.

2. Many chief complaint features are too rare to use on their own

Of the 200 chief complaint columns, 149 appear in less than 0.5% of encounters. These rare columns are unreliable as standalone predictors since there isn't enough data behind them to trust the patterns. **Planned response:** Group, exclude, or treat rare chief complaint features cautiously during modelling.

3. Possible leakage columns

Columns such as 'disposition' and 'previousdispo' may reflect information that wasn't actually available at the time of triage. A real triage model can only use what a nurse would know at the time of assessment. **Planned response:** Exclude any columns that might give the model information it wouldn't actually have at triage time.

Top 3 reasons to proceed

1. There's a real triage outcome to learn from

With 0 missing ESI labels, the model can be trained on the actual triage decisions made, not a rough substitute for them.

2. The key triage information is included in the data

The dataset includes clinically meaningful information available early in a visit such as vital signs, demographics, arrival details and chief complaints.

3. Early patterns line up with clinical expectations

Some early findings match what a clinician would expect. For example, chest pain is associated with more urgent ESI levels and oxygen saturation (a measure of how much oxygen is in the blood) has the closest link to how urgent a patient's case is, which makes sense clinically since low oxygen is often a sign of a serious problem.

Planned visual evidence

The final memo will use charts from the Week 5 data-quality dashboard, including:

- Missingness plot
- ESI and age distribution plot
- Race and ethnicity distribution plot
- Chief complaint frequency plot
- Vitals by ESI plot
- Vitals-and-ESI correlation heatmap
- Equity plot showing ESI distribution within each race category

Top-10 feature shortlist for the final memo

A ranked shortlist of features to test in Week 6:

1. triage_vital_o2 - Low oxygen levels are a warning sign for breathing problems and have the closest link to ESI of any vital sign.
2. triage_vital_rr - A faster breathing rate can be a sign of distress, infection, pain, or a lung problem.
3. triage_vital_hr - A higher heart rate can point to shock, pain, fever, anxiety, dehydration, or sudden illness.
4. triage_glucose - Abnormal blood sugar can signal that the body is unstable and may affect how urgent the case is.
5. cc_chestpain - Chest pain is a classic warning sign in the ED and was linked to more urgent cases.
6. cc_shortnessofbreath - Trouble breathing is urgent and linked to more urgent cases.
7. cc_suicidal - Thoughts of suicide often need urgent safety checks and were linked to more urgent cases.

8. cc_alteredmentalstatus - Sudden confusion or unresponsiveness is a major warning sign of serious illness or a brain-related problem.
9. cc_syncope - Fainting can point to a heart, nerve, or blood-flow problem and may need urgent assessment.
10. cc_abdominalpain - Abdominal pain is common in the ED and can range from minor illness to serious surgical or infection-related problems.

Caveats for the final memo

- Correlation shows a relationship, not proof that a feature causes higher or lower acuity.
- The Week 6 model is a first attempt, not something ready for hospital use.
- Very few patients are the most urgent level (ESI 1), so performance on the sickest patients needs separate testing.
- Many chief complaint columns are rarely used and may need to be grouped before modelling.
- Temperature is recorded in Fahrenheit, so fever/low-temperature cutoffs should use Fahrenheit units, not Celsius.
- Disposition-related variables should be excluded to avoid leakage, so the model isn't trained on information it wouldn't actually have at triage time.
- The model should be tested in the actual setting where it will be used. One ED's data may not represent every Caribbean hospital or patient population.